Technical Service Bulletin

Technical Service Bulletin: TSB150172	Released Date: 09-Dec-2015
Crankshaft Straightness Measurement	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- K38 (All Versions)
- K50 (All Versions)
- QSK38 (All Versions)
- QSK50 (All Versions)

This document outlines how to check crankshaft straightness.

Description of Method:

1.1. Crankshaft Measurement General Requirements

Measurement of crankshaft straightness with the specified accuracy can **only** be achieved if the crankshaft:

- Is correctly supported.
- Is at thermal equilibrium.
- Can be rotated smoothly.

These requirements can **not** be met by measuring the crankshaft while supported by bearing shells in a cylinder block. Rotation of the crankshaft with such an arrangement will lead to errors as the result of lateral and vertical displacement arising from curvature and clearance in the main bearings.

1.2. Measurement Environment

- Measurement **must** be conducted in a temperature controlled room at a temperature of 20°C [68°F] +/- 1 degree.
- The crankshaft to be measured **must** be stored in this room for at least twenty four hours to allow the crankshaft to stabilize at room temperature.
- Temperature of the crankshaft **must** be verified before measurements starts, and recorded on the measurement data worksheet included in this document. See measurement data worksheet section 1.8. below.

1.3. Measurement Setup

- The measurement table **must** be rated to support at least 450 kg [992 lb] and provide a stable, flat surface that allows V-blocks to be aligned and secured with the crankshaft centerline in a straight horizontal position.

The use of a granite measurement table and roller V-blocks are recommended. See Figure 1. Roller V-blocks have lower rolling resistance than standard V-blocks when rotating the crankshaft during measurement process.

Figure 1, Crankshaft on Roller V-blocks

- A motorized drive is preferred to be used to rotate the crankshaft smoothly over the full 360 degrees of rotation. Direct manual rotation **must** be done very carefully to provide the smooth motion necessary for consistent measurements.
- The height of V-blocks **must** be equal (Z-axis). The height difference between the V-blocks **must** be much smaller than part specified tolerance.

Example: If straightness specification is less than 0.06 mm [0.002 in] then the height difference **must** be 0.01 mm [0.0004 in] or less. The V-blocks height **must** be verified in the contact points of V-blocks with the crankshaft journal using a calibrated shaft or bar. Measurements **can** be made with a mechanical dial indicator or an electronic precision measurement system.

- The measuring device used **must** have a valid calibration certificate and **must** have resolution, repeatability, and accuracy (relationship of the real dimension to what the measurement device indicates) appropriate for the straightness specification being evaluated.

1.4. Measurement Procedure

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Use appropriately rated nylon sling and hoist to lift the crankshaft.
- Place the crankshaft onto roller V-blocks positioned to support the crankshaft on main journal number 2 and main journal number 8, as shown in Figure 2 below.
- Verify that the crankshaft is free **only** to rotate, with no vertical or lateral motion as the result of clearance in the rollers or rocking of the v-blocks.

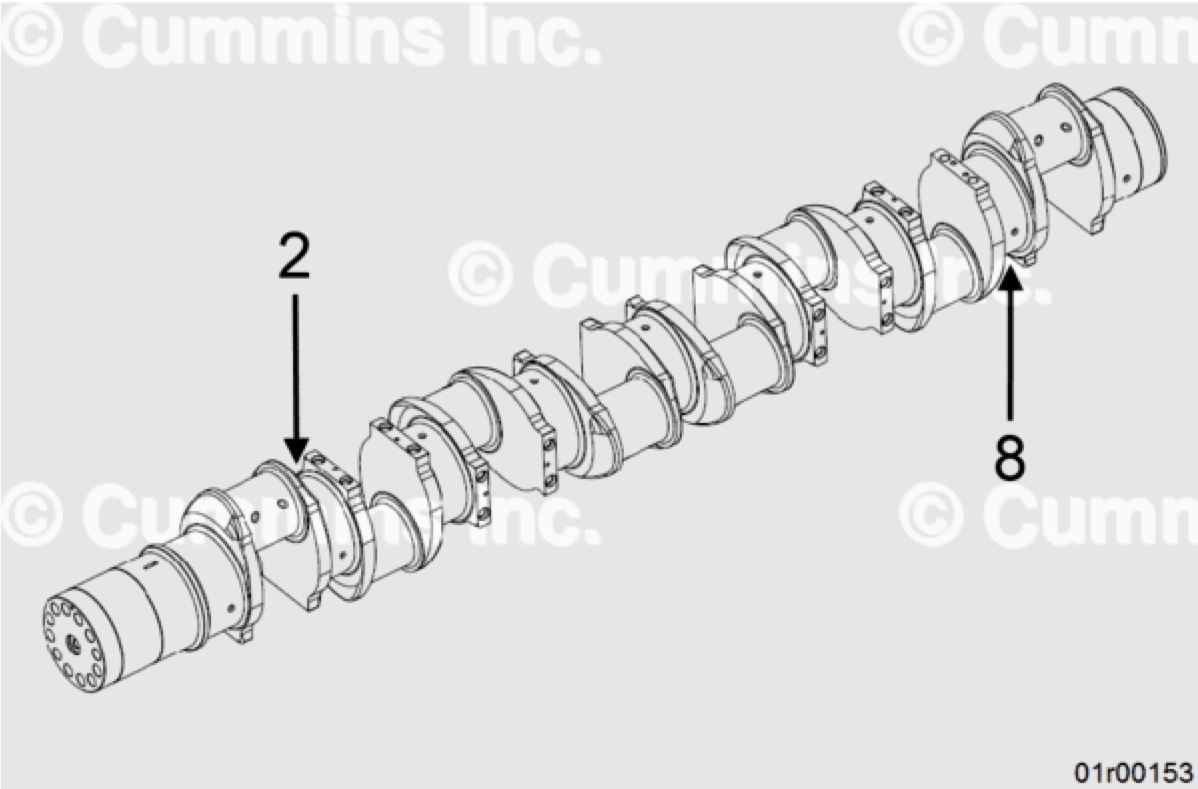

Figure 2, Main Bearing Journal Locations for Supporting Roller V-blocks.

- Using a marker pen, place a radial line on the front face of the crankshaft to align with top dead center (TDC) for crankshaft pin number one. This line represents the “hand” of the clock and indicates clock position of the crankshaft when viewed from the front.
- Position the dial indicator to enable measurement in the horizontal direction at the center line height of the crankshaft, as shown in Figure 3 below. Measurement readings **must** be rounded to the closest 0.001 mm [0.00004 in] and the angle of the high point of the main bearing journal runout **must** be determined to the nearest full hour clock position.

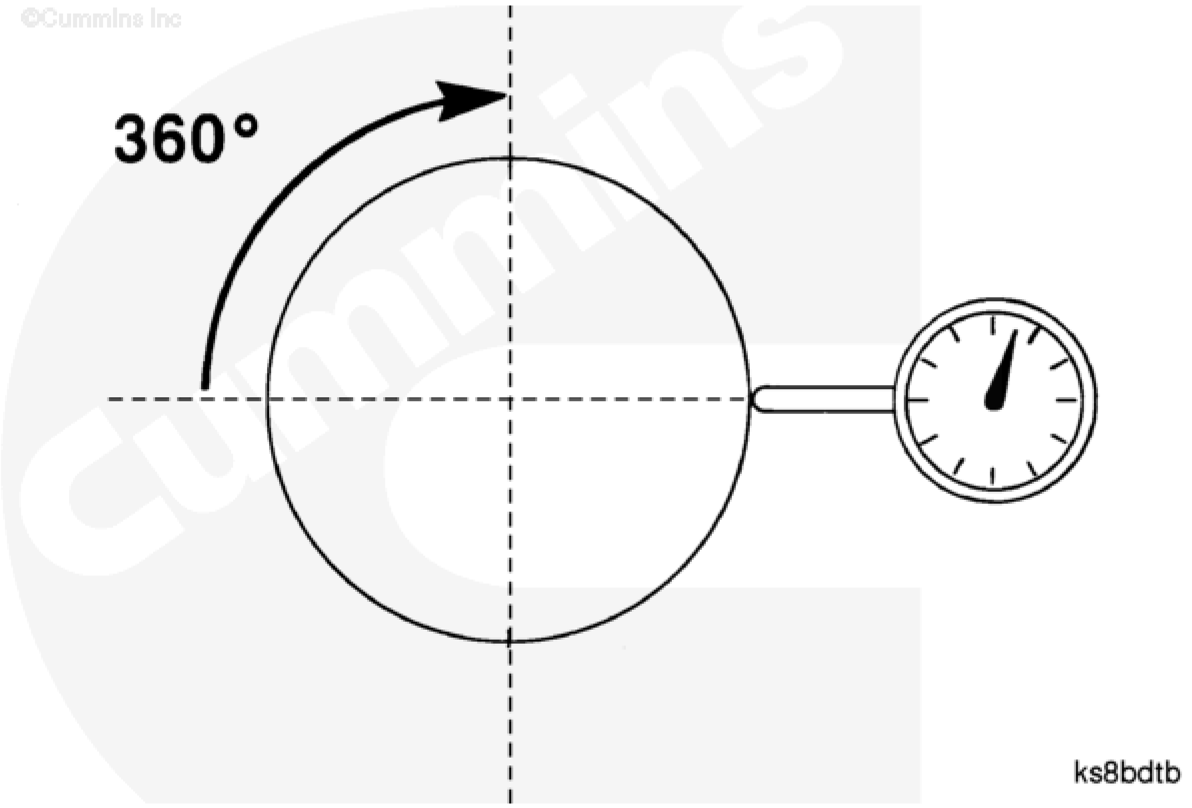

Figure 3, Position of Dial Indicator Stem on the Centreline of The Bearing Journal Being Measured.

- Measure in the middle of each main bearing journal, rotating slowly through two complete revolutions or 720 degrees.
- Read out the total indicator reading (TIR) and the clock position at which the maximum radial displacement occurs.

1.5. Determination of Crankshaft Straightness

- Record the total indicator reading (TIR) reading of each main bearing journal and the clock position at which the maximum displacement occurs. Determine if the crankshaft straightness is within specified limits.

1.6. Service Part Reuse Acceptance Levels

- Service acceptance levels for the crankshaft straightness, if **not** published, **must** be obtained from service engineering.

Table 1, All Main Bearing Journals Maximum Deflection Limit

	mm		**in**
K38, QSK38 Engine	0.203	MAX	0.0081
K50, QSK50 Engine	0.267	MAX	0.0105

Interpretation of the acceptance limits:

- Centerline position for all main bearing journals **must** lie within a cylinder of 0.203 mm [0.0081] diameter for K38 and QSK38 engines

- Centerline position for all main bearing journals **must** lie within a cylinder of 0.267 mm [0.0105] diameter for K50 and QSK50 engines

1.7. New Part Acceptance Levels

- New part acceptance levels for the crankshaft straightness, if **not** published, **must** be obtained from the crankshaft part drawing. To obtain the crankshaft part drawing, contact your local authorized Cummins® repair location.

Table 2, All Main Bearing Journals Maximum Deflection Limit

	mm		in
K38, QSK38 Engine	0.2	MAX	0.0079
K50, QSK50 Engine	0.254	MAX	0.01

Interpretation of the acceptance limits:

- Centerline position for all main bearing journals **must** lie within a cylinder of 0.200 mm [0.0079] diameter for K38 and QSK38 engines
- Centerline position for all main bearing journals **must** lie within a cylinder of 0.254 mm [0.0100] diameter for K50 and QSK50 engines

1.8. Measurement Data Worksheet

Measurement Data Worksheet

Date	12/3/2015	
Engine Model	QSK50	
Engine Serial Number (ESN)	87654321	
Hours	####	
Individual Journal Runout		
Journal (n)	**Measured Value (mm)**	**Clock position of Pin #1**
	R_n	C_n
1	0.000	0
2	0.020	8
3	0.060	8
4	0.070	3
5	0.100	4
6	0.070	8
7	0.060	8
8	0.020	8
9	0.000	0
	CTR = Crankshaft Total Runout (bow):	0.1
	MAXIMUM SPECIFICATION:	0.267 mm
	RESULT:	OK / NOT OK

Example Data

01r00154

Figure 4, Example Measurement Data

Calculation Steps:

1. Record the measured runout (R_n) and corresponding clock position (C_n) for each main journal in columns 2 and 3 by the corresponding journal number.

2. Crankshaft total runout (CTR) is the maximum R_n value recorded.

- If crankshaft total runout (CTR) is more than the maximum specification, then the part is out of specification
- If crankshaft total runout (CTR) is less than the maximum specification, then the part is within specification

Measurement Data Worksheet		
Date		
Engine Model		
Engine Serial Number (ESN)		
Hours		
Individual Journal Runout		
Journal (n)	Measured Value (mm)	Clock position of Pin #1
	R_n	C_n
1		
2		
3		
4		
5		
6		
7		
8		
9		
CTR = Crankshaft Total Runout (bow):		
MAXIMUM SPECIFICATION:		0.267 mm
RESULT:		OK / NOT OK

Document History

Date	Details
2015-10-27	Module Created
2015-12-8	Corrected title.

Last Modified: 09-Dec-2015